

Automated Multi-Modal Intoxication Detection Enclosure

Technical Architecture, Sensor Integration, and Tiered Diagnostic Logic

COMPREHENSIVE INTEGRATED SPECIFICATION

This document details the comprehensive technical specification, sensor orchestration, and advanced operational logic for a mask-based biometric screening device. By isolating the subject's sensory inputs and utilizing a tightly integrated array of physiological sensors, the system delivers an objective, automated, and legally defensible assessment of intoxication within a maximum 60-second operational window. This multi-layered architecture addresses the structural limitations of manual Standardized Field Sobriety Tests (SFSTs) by removing environmental variables and human observer bias.

1. System Architecture & Component Breakdown

The device consolidates seven distinct physiological and chemical sub-systems alongside an advanced vapor delivery mechanism into a unified edge-computing wearable enclosure:

- **Dual Infrared (IR) Cameras:** Independent, high-frame-rate sensors (minimum 90Hz) dedicated to tracking localized movements of both left and right eyes simultaneously.
- **Controlled Illumination Array:** A sealed infrared lighting matrix that ensures constant luminance profiles, eliminating ambient shadows, roadside glare, or variable lighting conditions.
- **High-Speed Eye Tracking Module:** Machine-vision algorithms processing real-time coordinates of the pupil center to map smooth pursuit, fixation stability, and saccadic velocity.
- **Pupillary Analysis Engine:** Software-driven measurement systems that continuously calculate static pupil diameter (down to 0.01mm) and dynamic constrictive/dilatory kinetics under light stimuli.
- **Integrated Microphone:** Acoustic sensor positioned within the oral cavity zone to capture verbal response latency, voice modulation, and articulate phoneme structural integrity.
- **Inertial Measurement Unit (IMU):** A multi-axis gyroscopic and accelerometer package mounted securely to the rigid chassis of the mask to monitor microscopic cranial sways, tremors, and vestibular oscillations.
- **Passive Alcohol Breath Sensor:** A high-accuracy fuel-cell or infrared chemical sensor placed directly in the respiratory exhaust line to capture tidal breathing samples.
- **Micro-Metered Vapor Delivery System:** Equipped with chemical-resistant fluoroelastomer seals, this system enables precise delivery of localized vapor challenges during active confirmatory testing phases.

2. Two-Stage Tiered Testing Logic

To optimize diagnostic accuracy and prevent unnecessary diagnostic intervention, the system operates on a hard-coded Two-Stage Tiered Testing Logic framework:

Stage 1: Passive Screening Phase (0s–45s)

This phase focuses on non-invasive monitoring of resting nystagmus, pupillary reflexes, speech biometrics, breath alcohol concentration, and head sway to form an initial biological baseline and risk score.

Stage 2: Active Confirmatory Challenge (+15s)

Triggered exclusively when Stage 1 identifies anomalies, conflicting physiological metrics, or indeterminate results, this stage introduces a controlled autonomic stressor to definitively confirm systemic impairment.

3. 60-Second Operational Timeline Matrix

The device executes an automated, sequential assessment pipeline over an exact 60-second window, governed by the following integrated timeline logic:

Time Interval	Sensor & System State	Diagnostic & Operational Function
00s – 10s	IMU Active; IR Cameras Calibrating	Enclosure sealing validation; baseline stabilization; structural registration of normal cranial movement and resting pupil dimensions.
10s – 35s	Controlled Illumination Active; Eye Tracking Engine Engaged	Automated Horizontal Gaze Nystagmus (HGN) screening via moving focal targets; micro-saccadic capture; smooth pursuit velocity logging.
35s – 45s	Variable Light Pulse; Microphone Triggered	Pupillary Light Reflex (PLR) challenge; latency calculations; automated vocal prompt and speech biometric capture.
45s – 55s	Conditional Active Challenge; Vapor Delivery Active (If Triggered)	Logic Fork: If Stage 1 metrics are clear, standard monitoring continues. If anomalies/conflicts are flagged, a Neurological Challenge micro-dose is actively administered.
55s – 60s	Chemical Sensor Sampling; Data Fusion & Purge Loop	Passive tidal breath collection & deep-lung equivalent calculation. Edge-computing data aggregation, multi-modal data fusion, and generation of the final mathematical impairment output. System purge initiated.

4. Sensor Cross-Correlation & Neurological Challenge Pharmacology

The diagnostic core of the system relies on advanced multi-modal data fusion rather than any single isolated metric. The central processing unit cross-correlates biometric data points to pinpoint central nervous system (CNS) depression or stimulation:

- **Oculomotor Integrity vs. Vestibular Balance:** The eye tracking modules map the physical tracking velocity of the eyes, while the IMU checks for minute vestibular corrections in head posture. When a substance compromises the cerebellum, the relationship between eye movement stabilization and physical sway exhibits a distinct mathematical divergence that cannot be manually faked or suppressed by the subject.
- **Neurological Challenge Pharmacology:** During the active challenge window, a sub-clinical micro-dose of volatile alkyl nitrite (**10–50 µg/L**) is introduced as an autonomic challenge. A sober, unimpaired autonomic nervous system rapidly compensates via baroreflex mechanisms. Conversely, an impaired system exhibits clear, measurable degradation across multiple vectors.

Additional Claimed System Outputs:

- Oculomotor instability metrics & pupillary response degradation profiles.
- Vestibular equilibrium disruption metrics tracking micro-spatial kinetics.
- A definitive, mathematical vector-based forensic output that logs raw coordinate shifts (**X, Y** pupil positions), pupillary reaction curves in milliseconds, breath concentration markers, and spatial kinetic frequencies.

5. Field Implementation Advantages

By moving the testing environment inside a controlled, face-mask-style enclosure, law enforcement and clinical agencies resolve critical operational vulnerabilities:

1. **Environmental Insulation:** Traditional roadside or tactical environments are subject to wind, dust, temperature fluctuations, and distracting visual fields. The enclosure standardizes the testing environment globally, ensuring identical testing parameters regardless of external conditions.
2. **Elimination of Passive Co-operation Barriers:** Conventional chemical breath testing requires active, sustained exhalation volumes from the subject. The mask architecture captures normal tidal breathing pattern loops seamlessly, drastically lowering the collection failure rate caused by user uncooperativeness or physical respiratory fatigue.